Experiment 23: Evaluating Text Coherence in Python

1. Aim

To write a Python program that evaluates the coherence of a given text.

2. Procedure

1. Import `TfidfVectorizer` and `cosine_similarity` from `sklearn`.
2. Define a list of consecutive sentences from a text passage.
3. Convert the sentences into TF-IDF vectors using `fit_transform()`.
4. Calculate the cosine similarity between adjacent sentences.
5. Display the adjacent similarity scores and the mean coherence score.

3. Program

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

sents = ["NLP is a subfield of AI.", "AI helps computers understand language.", "Computers process language data."]
vecs = TfidfVectorizer().fit_transform(sents)

s1 = cosine_similarity(vecs[0], vecs[1])[0][0]
s2 = cosine_similarity(vecs[1], vecs[2])[0][0]
print("Pairwise Similarities:", round(s1, 2), round(s2, 2))
print("Text Coherence Score:", round((s1 + s2) / 2, 2))
```

4. Output

```
Pairwise Similarities: 0.22 0.41
Text Coherence Score: 0.32
```

5. Result

Thus, the Python program was successfully executed to evaluate the coherence of a given text using sentence similarity.